

Mohammad Masum, PhD.

San Jose, CA

470-900-8624

mohammad.masum@sjsu.edu

[Google Scholar](#)

EDUCATION

- | | |
|------|--|
| 2021 | Ph.D. in Analytics and Data Science, Kennesaw State University, Kennesaw, GA |
| 2017 | MS in Mathematical Sciences, East Tennessee State University, Johnson City, TN |
| 2014 | MS in Applied Mathematics, University of Dhaka, Dhaka, Bangladesh |
| 2012 | BS in Mathematics, University of Dhaka, Dhaka, Bangladesh |

PROFESSIONAL APPOINTMENTS

- | | |
|------|--|
| 2022 | Assistant Professor, Department of Applied Data Science, San Jose State University |
| 2021 | Postdoctoral Researcher, Department of Information Technology, Kennesaw State University |
| 2018 | Data Science Research Analyst (Intern), GE Power |

PUBLICATIONS

Peer-Reviewed Journal Articles

Yu, C. H., & **Masum, M.** (2026). HybridSense-LLM: A Structured Multimodal Framework for Large-Language-Model–Based Wellness Prediction from Wearable Sensors with Contextual Self-Reports. *Bioengineering*, 13(1), 120. **Impact Factor: 3.7**

Sreenivas, S. C., Chowdhury, S., & **Masum, M.** (2025). Enhancing Clinical Named Entity Recognition via Fine-Tuned BERT and Dictionary-Infused Retrieval-Augmented Generation. *Electronics*, 14(18), 3676. <https://doi.org/10.3390/electronics14183676>. **Impact Factor: 2.6**

Balachandran, A., & **Masum, M.** (2025). A multimodal framework for enhancing E-commerce information management using vision transformers and large language models. *International Journal of Information Management Data Insights*, 5(2), 100355. **Cite Score 26.5**

Basavaraju, B. K., & **Masum, M.** (2025). LSE-Net: Integrated Segmentation and Ensemble Deep Learning for Enhanced Lung Disease Classification. *Electronics*, 14(12), 2407. **Impact Factor: 2.6**

Devadasan, A. V., Sengupta, S., & **Masum, M.** (2025). Deep Learning for Non-Invasive Blood Pressure Monitoring: Model Performance and Quantization Trade-Offs. *Electronics*, 14(7), 1300. **Impact Factor: 2.6**

Gupta, M., Dheekonda, V., & **Masum, M.** (2024). Genie: Enhancing information management in the restaurant industry through AI-powered chatbot. *International Journal of Information Management Data Insights*, 4(2), 100255. **Cite Score 26.5**

Lowhorn, R. J., Chowdhury, M., Kimitei, S., Haskin, S., **Masum, M.**, & Rahman, A. F. (2024). Comorbidities and their association with COVID-19 mortality in Mexico between January 2020 and August 2021. *PLOS ONE*, 19(4), e0296895. **Impact Factor: 2.9**

Masum, M., Masud, M A., Adnan, M., Shahriar, H., Kim, S. (2021). Comparative Study of a Mathematical Epidemic Model, Statistical Modeling, and Deep Learning for COVID-19 Forecasting and Management. *Socio-economic Planning Sciences*. **Impact Factor: 4.9**

Zahangir, S., Chowdhury, M., Khan, H., **Masum, M.**, (2021). Trends and determinants of knowledge and awareness of HIV/AIDS among married women in Bangladesh: An urban-rural comparison. *Demografie, Review for Population Research*.

Masum, M., Shahriar, H., Haddad, H. (2021). A Transfer Learning with Deep Neural Network Approach for Network Intrusion Detection. *International Journal of Intelligent Computing Research (IJICR)*. **Impact Factor: 6.6**

Peer-Reviewed Conference Papers/Proceedings

Dunton, A., Govindarajulu, P., & **Masum, M.** (2025). *Do multimodal large language models compete with specialized object detection models? A quantitative benchmark*. In Proceedings of the 2025 IEEE International Conference on Big Data. IEEE. **Acceptance Rate: 18.7%. H5 Index: 55**

Yadav, T., & **Masum, M.** (2026). *HRIT: A human-readable framework for phishing URL detection using large language models*. In Proceedings of the 2026 IEEE 5th International Conference on AI in Cybersecurity (ICAIC). IEEE.

Dunton, A., Vyas, R., Shah, H., Sankathala, S., & **Masum, M.** (2025, May). InvadeAI: INteractive ADvertising with Multimodal AI. In *2025 IEEE Conference on Artificial Intelligence (CAI)* (pp. 126-131). IEEE.

Balachandran, A., & **Masum, M.** (2025, May). Visiorag: A Multimodal Framework for Enhancing Recommendation System Using Vision Transformers and Rag. In *2025 IEEE Conference on Artificial Intelligence (CAI)* (pp. 114-119). IEEE.

Suryadevara, L. S. R., Lagadapati, N., Velamala, V. S. K., Potti, Y. L. S., & **Masum, M.** (2025, May). BARTNet: Context-Aware Deep Learning Framework for BART Ridership Forecasting. In *2025 IEEE Conference on Artificial Intelligence (CAI)* (pp. 825-828). IEEE.

Arcot, S., **Masum, M.**, Kader, M. S., & Saha, A. (2024). TinyML for cybersecurity: Deploying optimized deep learning models for on-device threat detection on resource-constrained devices. *IEEE International Conference on Big Data (Big Data 2024)*.

Veerla, S., Devadasan, A. V., **Masum, M.**, Chowdhury, M., & Shahriar, H. (2024, July). E-SMOTE: Entropy Based Minority Oversampling for Heart Failure and AIDS Clinical Trails Analysis. In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)* (pp. 1841-1846). IEEE.

Bhyregowda, P., **Masum, M.**, Mamudu, L., Chowdhury, M., Kosaraju, S. C., & Shahriar, H. (2024, July). ActivePCA: A Novel Framework Integrating PCA and Active Machine Learning for Efficient Dimension Reduction. In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)* (pp. 320-325). IEEE.

Rahman, M. A., Akter, M. S., Miller, E., Timofti, B., Shahriar, H., **Masum, M.**, & Wu, F. (2024, July). Fine-tuned Variational Quantum Classifiers for Cyber Attacks Detection based on Parameterized Quantum Circuits and Optimizers. In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)* (pp. 1067-1072). IEEE.

Gupta, M., Shah, S., **Masum, M.**, & Kosaraju, S. C. (2024, January). CMI: Cluster-Centric Missing Value Imputation with Feature Consistency. In *2024 IEEE 14th Annual Computing and Communication Workshop and Conference (CCWC)* (pp. 0521-0526). IEEE. **Acceptance Rate 38.7%**

Jobair, F., Shashank, T., **Masum, M.**, Maria, V., Shahriar, H., & Sheikh, A. (2022). A Novel IoT based Framework for Non-Invasive Human Hygiene Monitoring using Machine Learning Technique. *10th IEEE International Conference on Healthcare Informatics. IEEE.*

Masum, M., Anjum, N., Faruk, M. J. H., Shahriar, H., Sakib, N., Valero, M., ... & Cuzzocrea, A. (2022, December). A Novel Machine Learning Based Framework for Bridge Condition Analysis. In *2022 IEEE International Conference on Big Data (Big Data)* (pp. 5530-5535). IEEE.

Akter, M. K., Faruk, M. J. H., Anjum, N., **Masum, M.**, Shahriar, H., Sakib, N., Rahman, A., Wu, F., & Cuzzocrea, A. (2022). Software Supply Chain Vulnerabilities Detection in Source Code: Performance Comparison between Traditional and Quantum Machine Learning Algorithms. In *2022 IEEE International Conference on Big Data (Big Data)* (pp. 5639-5645).

Masum, M., H., Jobair, H., Shahriar, H., Qian, K., Lo, D., and Adnan, M. (2022). Ransomware Classification and Detection with Machine Learning Algorithms. In *2022 IEEE Annual Computing and Communication Workshop and Conference. IEEE.*

Jobair, H., Miner, P., Coughlan, R., **Masum, M.**, Shahriar, H., Clincy, V., Centinkaya, C. (2021). Smart Connected Aircraft: Towards Security, Privacy, and Ethical Hacking. *International Conference on Security of Information and Networks (SINCONF 2021). IEEE.*

Masum, M., Shahriar, H., Haddad, H., & Wenzhan Song (2021, September). A Statistical Summary Analysis of Window-Based Extracted Features for EEG Signal Classification. In *2021 IEEE International Conference for Digital Health. IEEE. Acceptance Rate: 20%*

Masum, M., Shahriar, H. (2020, December). TL-NID: Deep Neural Network with Transfer Learning for Network Intrusion Detection. In *2020 IEEE International Conference for Internet Technology and Secured Transactions*. IEEE. **Acceptance Rate: 15%**

Masum, M., Shahriar, H., Haddad, H., Jobair, H., Rahman, Adnan, M., Cuzzocrea, A., Khan, A., Valero, M., and Wu, F. (2021). Bayesian Hyperparameter Optimization for a Deep Neural Network Based Network Intrusion Detection. In *2021 IEEE International Conference on Big Data (Big Data)*. IEEE

Masum, M., Shahriar, H., Haddad, H., Ahamed, S., Sneha, S., Rahman, M., & Cuzzocrea, A. (2020, December). Actionable Knowledge Extraction Framework for COVID-19. In *2020 IEEE International Conference on Big Data (Big Data)*. IEEE.

Masum, M., Shahriar, H., Haddad, H., & Alam, M. (2020, December). r-LSTM: Time Series Forecasting for COVID-19 Confirmed Cases with LSTM-based Framework. In *2020 IEEE International Conference on Big Data (Big Data)*. IEEE. **Acceptance Rate: 16%**

Masum, M., Shahriar, H., & Haddad, H. M. (2020, July). Epileptic Seizure Detection for Imbalanced Datasets Using an Integrated Machine Learning Approach. In *2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)* (pp. 5416-5419). IEEE. **Acceptance Rate: 12%**

Masum, M., Shahriar, H., & Haddad, H. (2020, July). Analysis of Sampling Techniques Towards Epileptic Seizure Detection from Imbalanced Dataset. In *2020 IEEE 44th Annual Computers, Software, and Applications Conference (COMPSAC)* (pp. 684-692). IEEE. **Acceptance Rate: 24%**

Li Z., Zhang H., **Masum M.**, Shahriar H., Haddad H. Cyber Fraud Prediction with Supervised Machine Learning Techniques. *ACM Southeast Conference – ACMSE 2020*, ISBN: 978-1-4503-7105-6 Tampa, FL, USA, April 2-4, 2020

Masum, M., Shahriar, H. (2019, December). Droid-NNet: Deep Learning Neural Network for Android Malware Detection. In *2019 IEEE International Conference on Big Data (Big Data)* (pp. 5789-5793). IEEE.

Tsaku, N. Z., Kosaraju, S. C., Aqila, T., **Masum, M.**, Song, D. H., Mondal, A. M., ... & Kang, M. (2019, November). Texture-based Deep Learning for Effective Histopathological Cancer Image Classification. In *2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 973-977). IEEE.

Kosaraju, S. C., **Masum, M.**, Tsaku, N. Z., Patel, P., Bayramoglu, T., Modgil, G., & Kang, M. (2019, September). DoT-Net: Document Layout Classification Using Texture-Based CNN. In *2019 International Conference on Document Analysis and Recognition (ICDAR)* (pp. 1029-1034). IEEE. **Acceptance Rate: 13%**

Hao, J., **Masum, M.**, Oh, J. H., & Kang, M. (2019, June). Gene-and Pathway-Based Deep Neural Network for Multi-omics Data Integration to Predict Cancer Survival Outcomes. In *International Symposium on Bioinformatics Research and Applications* (pp. 113-124). Springer, Cham. **Acceptance Rate: 20%**

Masum, M., Kosaraju, S., Bayramoglu, T., Modgil, G., & Kang, M. (2018, October). Automatic knowledge extraction from OCR documents using hierarchical document analysis. In *Proceedings of the 2018 Conference on Research in Adaptive and Convergent Systems* (pp. 189-194). ACM. **Acceptance Rate: 24%**

Manuscripts under Review

Kanchugantla, Y., & Masum, M. (2026). Complexity-aware in-context retrieval for executable analytics over tabular data. *ACM Transactions on Information Systems*. Manuscript under review.

Sun, Y., & Masum, M. (2026). PE2Prompt: A large language model-native framework for interpretable static malware analysis. *Knowledge-Based Systems*. Manuscript under review.

AWARDS AND HONORS

2024

Faculty Mentor, SJSU Student RSCA Competition Winner. *A Novel Framework Integrating PCA and Active Machine Learning for Efficient Dimension Reduction*. Student: Priyanka Bhyregowda

2022

Best Paper Award, *Tenth IEEE International Conference on Healthcare Informatics*, IEEE, USA, for *A Novel IoT-based Framework for Non-Invasive Human Hygiene Monitoring using Machine Learning Techniques*.

2020

Symposium of Student Scholars Research Award, Kennesaw State University, USA, for *r-LSTM: Time Series Forecasting for COVID-19 Confirmed Cases with LSTM-based Framework*. First Runner-up, Graduate Category.

GRANTS AND FELLOWSHIPS

2025

PI, CURF ASPIRE Grant (Advancing Student Projects in Research Excellence), College of Information, Data and Society, San José State University, USA. *Device-Agnostic Wellness Prediction: Evaluating LLM Robustness to Wearable Sensor Heterogeneity*. \$4,500.

2025

Participant, Research Impact Academy, San José State University, USA. Award #25-RIA-07-001. \$2,000.

2025

Faculty Mentor, R&I Student RSCA Fellowship, San José State University, USA. *Dual-Phase Study on LLM-Based Malware Creation and Defense*. Student: Tanya Yadav. \$3,000.

2024

PI, SJSU RSCA Seed Grant, San José State University, USA. *TinyPulse+: Bridging Gaps in Blood Pressure Monitoring for a Healthier Tomorrow*. \$7,500.

INVITED TALKS

2022

Introduction to Research Methods & Manuscript Preparation' at California State University, Fullerton. November 5, 2022

CONFERENCE ACTIVITIES

Program Committee Members

2024 COMPSAC (Smart & Connected Health)

2024 IEEE ICDH

2024 IEEE ICDH

2023 COMPSAC (The 17th IEEE International Workshop on Security, Trust, and Privacy for Software Applications, Smart & Connected Health, and Invited Papers)

2022 COMPSAC (The 16th IEEE International Workshop on Security, Trust, and Privacy for Software Applications, Student Research Symposium, and Invited Papers)

2022 IEEE ICDH

2021 COMPSAC (Smart & Connected Health)

2019 ACMSE (Big Data, Data Mining, HPC, and Machine Learning Track)

Session Chair

2024 IEEE COMPSAC

Paper presented

InvaDeAI: Interactive ADvertising with Multimodal AI. IEEE CAI
May 5-7, 2025, CA, USA

BARTNet: Context-Aware Deep Learning Framework for BART Ridership Forecasting. IEEE CAI
May 5-7, 2025, CA, USA

E-SMOTE: Entropy Based Minority Oversampling for Heart Failure and AIDS Clinical Trails Analysis, IEEE COMPSAC
July 2-4, 2024, Osaka, Japan

ActivePCA: A Novel Framework Integrating PCA and Active Machine Learning for Efficient Dimension Reduction, IEEE COMPSAC
July 2-4, 2024, Osaka, Japan

TL-NID: Deep Neural Network with Transfer Learning for Network Intrusion Detection, IEEE 15th International Conference on Internet Technology and Secured Transactions (ICITST) December 8-10, 2020, London, United Kingdom

r-LSTM: Time Series Forecasting for COVID-19 Confirmed Cases with LSTM-based Framework, IEEE International Conference on Big Data December 10-13, 2020, Virtual

Actionable Knowledge Extraction Framework for COVID-19, IEEE International Conference on Big Data December 10-13, 2020, Virtual

Epileptic Seizure Detection for Imbalanced Datasets Using an Integrated Machine Learning Approach, 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) July 20-24, 2020, Virtual

Analysis of Sampling Techniques Towards Epileptic Seizure Detection from Imbalanced Dataset, IEEE COMPSAC July 13-17, 2020, Virtual

Cyber Fraud Prediction with Supervised Machine Learning Techniques. ACM Southeast Conference April 2-4, 2020, FL, USA

Gene-and Pathway-Based Deep Neural Network for Multi-omics Data Integration to Predict Cancer Survival Outcomes, International Symposium on Bioinformatics Research and Application June 3-6, 2019, Barcelona, Spain

TEACHING EXPERIENCE

San Jose State University
Machine Learning Technologies
Spring 2026

Generative Model Applications
Spring 2026, Fall 2025, Spring 2025

Deep Learning Technologies
Fall 2025, Spring 2025, Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2022

Advanced Mathematical Methods for Data Intelligence
Spring 2026, Fall 2025, Spring 2025

Mathematical Methods for Data Analytics
Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2022

Kennesaw State University

Elementary Statistics

Spring 2021

East Tennessee State University

Probability and Statistics – Non-calculus

Fall 2016, Spring 2017

TEACHING & PROFESSIONAL DEVELOPMENT

2026

Participant, Building a Thriving Faculty (Community of Practice),
San José State University, USA.

2026

Selected Participant, TILting Towards Equity (Transparency in Learning and Teaching),
Center for Faculty Excellence and Teaching Innovation,
San José State University, USA.

2022

Badge Recipient, ACUE Module: Checking for Student Understanding in Online Learning,
Association of College and University Educators (ACUE).

SERVICE TO PROFESSION**Judge**

2025

Served as Judge for IBM SkillsBuild Hackathon 2025, Reviewed and scored Agentic AI-driven innovation projects.

Guest Editor

2025

Lead Guest Editor for a Special Issue of Bioengineering, MDPI journal, titled – “*Multimodal Generative AI for Health Monitoring and Bioengineering Applications*”

2024

Lead Guest Editor for a Special Issue of Electronics, MDPI journal, titled – “*Artificial Intelligence and Big Data Processing for Healthcare*”. This issue focuses on the integration of AI and Big Data to transform healthcare systems and processes.

Manuscript and Peer Reviewer

2020–Present

Reviewed manuscripts for various journals and conferences, including Frontiers, Biosensors, Future Internet, Information, Computers, Sustainability, Algorithms, Sustainability, IEEE ICDH 2024, IEEE COMPSAC 2024, Bulletin of Electrical Engineering and Informatics, IEEE BIBM, IEEE Access, ISBRA, NSys 2020, and IGI Global.

DEPARTMENT/UNIVERSITY SERVICE**Department AI Ambassador, San José State University,**

Spring–Summer 2026

Selected to represent the Applied Data Science department in campus initiatives on responsible AI integration in teaching and learning.

Student Evaluation Review Board (SERB)

2026–Present

Serve as faculty representative on University-wide board overseeing student evaluation instruments.

Search Committee, Associate Dean for Faculty Success, Research, and Industry Partnerships

2026

Serve on college-level search committee focused on faculty success, research, and industry partnerships.

CIDS RSCA Advisory Committee (RAC)

2025–Present

Serve on the College of Information, Data & Society RSCA Advisory Committee, providing input on research initiatives, supporting faculty and student scholarship, and contributing to the annual CIDS Student Research Conference.

Graduate Research Advisor, Department of Applied Data Science

2025–Present

Coordinate MS thesis activities including student advising, supervision support, progress tracking, and reporting. Lead workshops and guide students on thesis pathways.

General Education (GE) Summer Workgroup

2024

Participated in drafting revisions to University GE policies and guidelines, collaborating with colleagues across colleges and providing feedback for future GE initiatives.

Scholarship Review Committee

2024–2025

Reviewed and evaluated scholarship applications for the academic year, ensuring a fair selection process for university-wide scholarships.

Thesis Supervision

2024–Present

Committee Chair for eight MS students in Applied Data Science at San José State University.

Thesis Committee Membership

2022–Present

Committee Member for seven MS students in Applied Data Science and Computer Science departments.

Departmental Level Committees

2022–Present

Contributed to the development and implementation of academic policies and initiatives within the department.

Equity, Diversity, Inclusion, and Social Justice (EDISJ) Committee

2022–2025

Supported and contributed to efforts promoting diversity, equity, and inclusion, fostering social justice initiatives in the academic community.